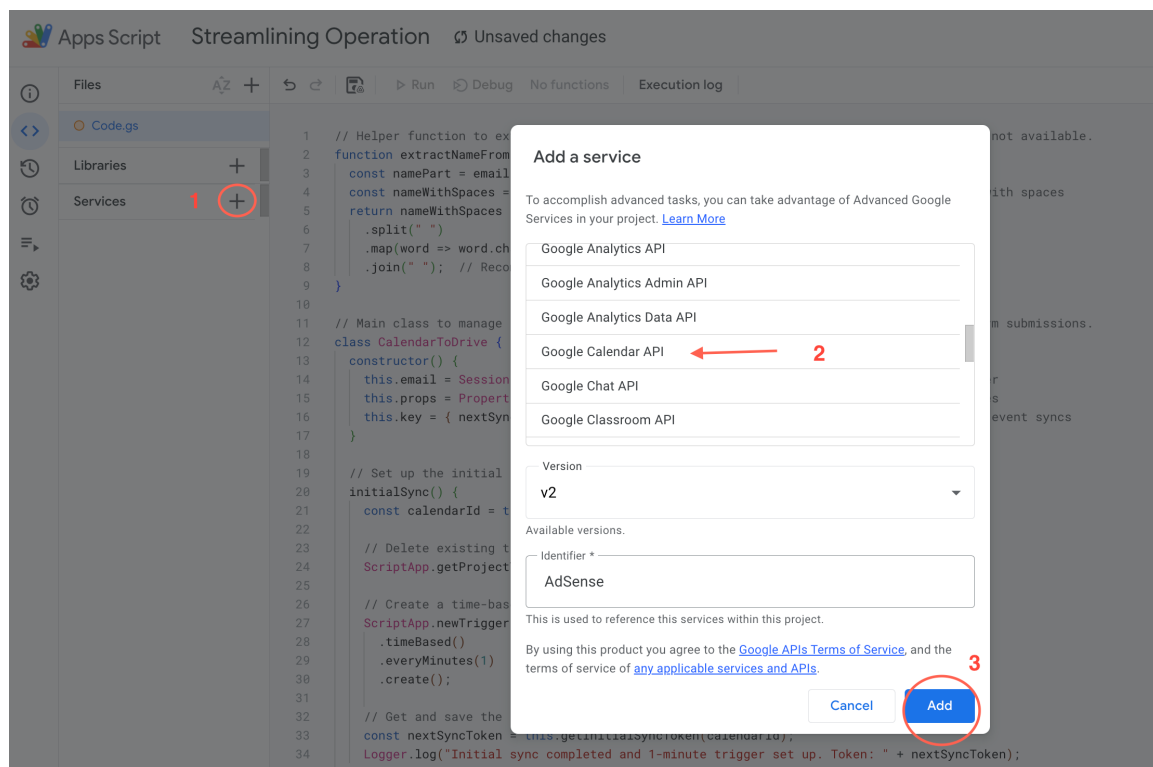


Implementation Guide: Google Apps Script Case Study

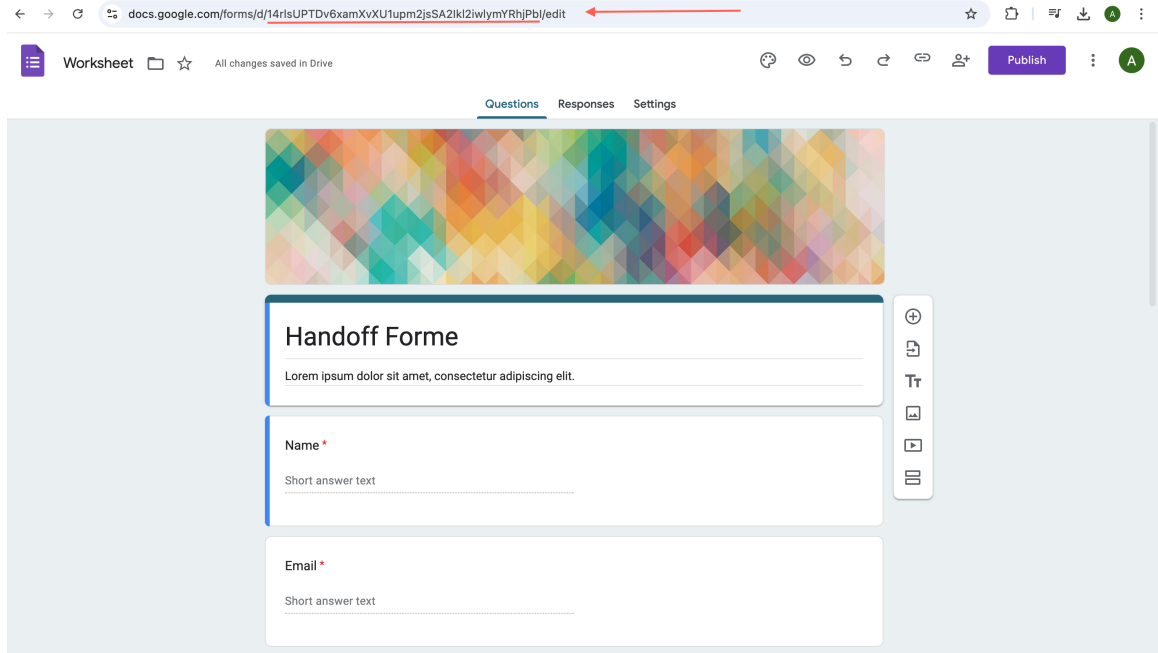
Step 1: Setup the Script

1. Open Google Apps Script by visiting script.google.com
2. Create a new project and copy-paste the provided Google Apps Script code.
3. Enable the Google Calendar API (Advanced Service):
 - Go to Services (gear icon) > + Add a service and add Google Calendar.



Step 2: Update the Script with the Correct Form ID

- Open your Google Form and navigate to the edit page.
- The edit URL will look like this:
“<https://docs.google.com/forms/d/14rlsUPTDv6xamXvXU1upm2jsSA2Ikl2iwlymYRhjPbI/edit>”
- The Form ID is the part of the URL between /d/ and /edit. For example, in the above URL, the Form ID is :
“14rlsUPTDv6xamXvXU1upm2jsSA2Ikl2iwlymYRhjPbI”



- Replace the form ID in the function `createFormTrigger()`:

```
const formId = "YOUR_FORM_ID"; // Replace with your form's unique ID
```

Step 3: Initial Setup (Run Manually Once)

During the initial setup and before setting up your triggers, you'll be asked to grant the following permissions:

1. Access Google Calendar: To read and sync calendar events.
2. Access Google Drive: To create folders and save PDF files.
3. Access Google Forms: To retrieve form responses.
4. Access Gmail: To send email notifications to participants.

How to Accept Permissions:

1. When you run `initialSync()` or `createFormTrigger()` for the first time, you will be prompted with a Google authorization screen.
2. Select your Google account and click Allow to grant the necessary permissions.

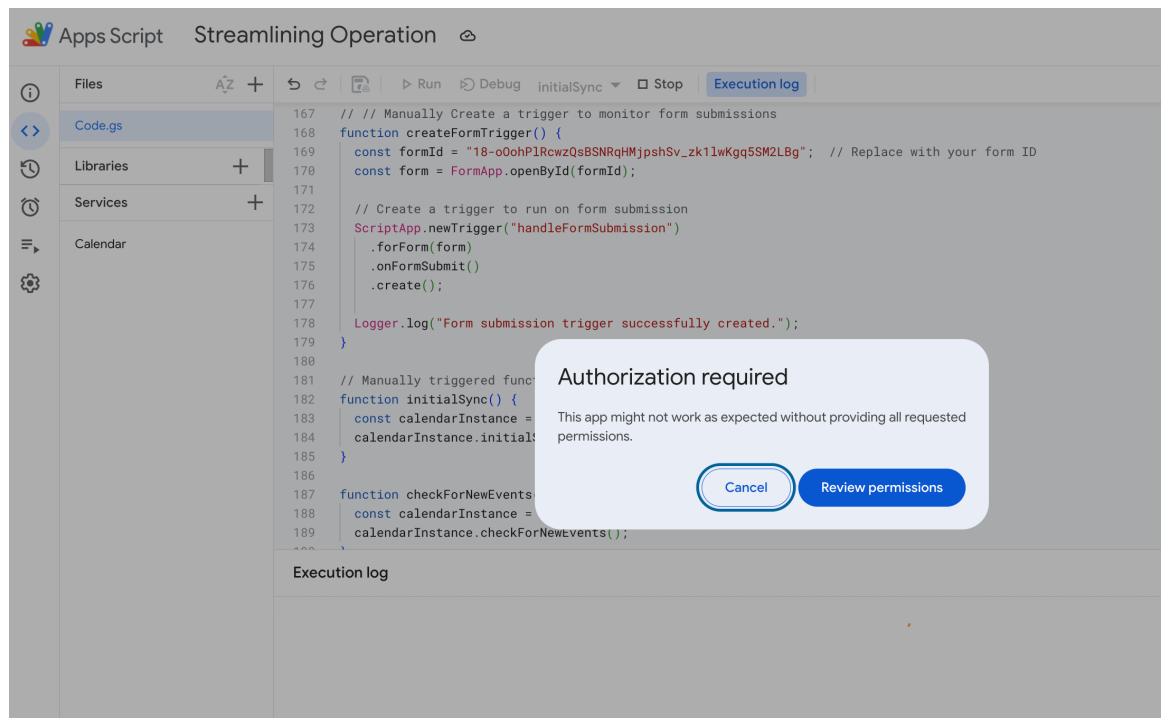
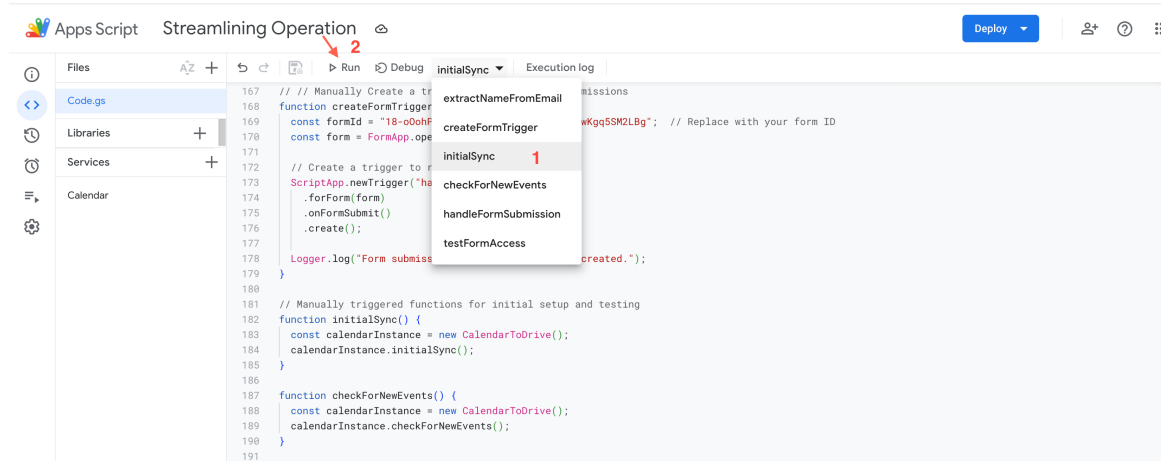
Run `initialSync()` and `createFormTrigger()` Manually Once

1. Run `initialSync()` manually:
 - Sets up the time-based trigger to check for new Google Calendar events every

minute.

2. Run createFormTrigger() manually:

- Creates the trigger to monitor form submissions and handle them automatically.



Streamlining Operation wants access to your Google Account

 asmahass@usc.edu

Select what **Streamlining Operation** can access

☒ Select all

 Read, compose, send, and permanently delete all your email from Gmail. [Learn more](#) ☒

 See, edit, create, and delete all of your Google Drive files. [Learn more](#) ☒

 See the events on Google calendars you own. [Learn more](#) ☒

☐ View and manage your forms in Google Drive. [Learn more](#) ☒

 Allow this application to run when you are not present. [Learn more](#) ☒

Because you're using Sign in with Google, Streamlining Operation will be able to

 Associate you with your personal info on Google ☒

 See your primary Google Account email address ☒

Make sure you trust Streamlining Operation

 [Learn why you're not seeing links to Streamlining Operation's Privacy Policy or Terms of Service](#)

Review Streamlining Operation's Privacy Policy and Terms of Service to understand how Streamlining Operation will process and protect your data.

To make changes at any time, go to your [Google Account](#).

Learn how Google helps you [share data safely](#).

Cancel

Continue

Apps Script Streamlining Operation

Deploy

Files

Code.gs

Libraries

Services

Calendar

```
167 // // Manually Create a trigger to monitor form submissions
168 function createFormTrigger() {
169   const formId = "18-o0ohP1RwzQsBSNRqHwjshSv_zk1lwKg5SM2LBg"; // Replace with your form ID
170   const form = FormApp.openById(formId);
171
172   // Create a trigger to run on form submission
173   ScriptApp.newTrigger("handleFormSubmission")
174     .forForm(form)
175     .onFormSubmit()
176     .create();
177
178   Logger.log("Form submission trigger successfully created.");
179 }
180
181 // Manually triggered functions for initial setup and testing
182 function initialSync() {
183   const calendarInstance = new CalendarToDrive();
184   calendarInstance.initialSync();
185 }
186
187 function checkForNewEvents() {
188   const calendarInstance = new CalendarToDrive();
189   calendarInstance.checkForNewEvents();
190 }
```

Execution log

4:50:24 PM	Notice	Execution started
4:50:30 PM	Info	Next sync token saved.
4:50:30 PM	Info	Initial sync token saved.
4:50:30 PM	Info	Initial sync completed and 1-minute trigger set up. Token: COCanIywsYsDEOCanIywsYsDGAUgjrnS2AiojrnS2AI=
4:50:30 PM	Notice	Execution completed

Apps Script Streamlining Operation

Deploy

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Calendar

```
167 // // Manually Create a tr
168 function createFormTrigger
169   const formId = "18-o0ohP
170   const form = FormApp.ope
171
172   // Create a trigger to r
173   ScriptApp.newTrigger("ha
174     .forForm(form)
175     .onFormSubmit()
176     .create();
177
178   Logger.log("Form submiss
179   created.");
180
181 // Manually triggered functions for initial setup and testing
182 function initialSync() {
183   const calendarInstance = new CalendarToDrive();
184   calendarInstance.initialSync();
185 }
186
187 function checkForNewEvents() {
188   const calendarInstance = new CalendarToDrive();
189   calendarInstance.checkForNewEvents();
190 }
```

createFormTrigger

- extractNameFromEmail
- createFormTrigger
- initialSync
- checkForNewEvents
- handleFormSubmission
- testFormAccess

The screenshot displays the Google Apps Script editor interface. The top bar shows 'Apps Script' and the project name 'Streamlining Operation'. The left sidebar contains a 'Files' panel with 'Code.gs' selected, and a 'Services' panel with 'Calendar' selected. The main editor area shows the 'Code.gs' file with the following JavaScript code:

```
167 // // Manually Create a trigger to monitor form submissions
168 function createFormTrigger() {
169   const formId = "18-o0ohP1RcwzQsBSNRqHMjpshSv_zk1lwKgq5SM2LBg"; // Replace with your form ID
170   const form = FormApp.openById(formId);
171
172   // Create a trigger to run on form submission
173   ScriptApp.newTrigger("handleFormSubmission")
174     .forForm(form)
175     .onFormSubmit()
176     .create();
177
178   Logger.log("Form submission trigger successfully created.");
179 }
180
181 // Manually triggered functions for initial setup and testing
182 function initialSync() {
183   const calendarInstance = new CalendarToDrive();
184   calendarInstance.initialSync();
185 }
186
187 function checkForNewEvents() {
188   const calendarInstance = new CalendarToDrive();
189   calendarInstance.checkForNewEvents();
190 }
```

Below the code editor, the 'Execution log' is visible, showing three entries:

Time	Message	Status
4:57:09 PM	Execution started	Notice
4:57:11 PM	Form submission trigger successfully created.	Info
4:57:11 PM	Execution completed	Notice

The bottom section of the screenshot shows the 'Triggers' panel. It contains a table with the following data:

Owned by	Last run	Deployment	Event	Function	Error rate
Me	Feb 7, 2025, 4:58:16 PM	Head	Time-based	checkForNewEvents	0%
Me	-	Head	From form - On form submit	handleFormSubmission	-

Step 4: How It Works

Step 4.1: When a Google Calendar event is created:

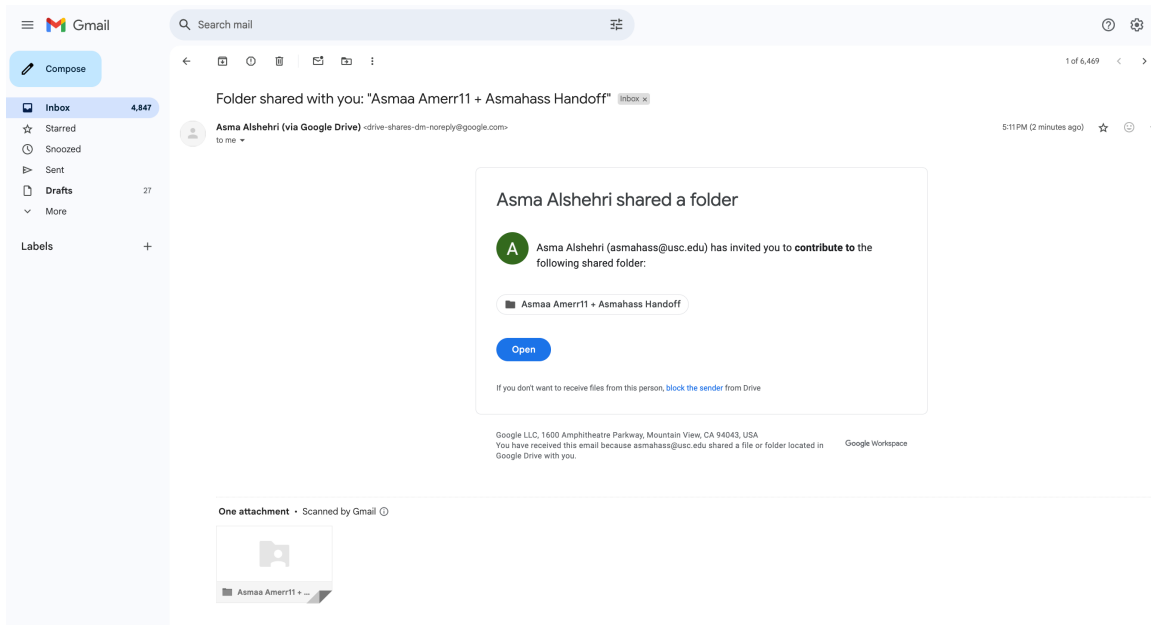
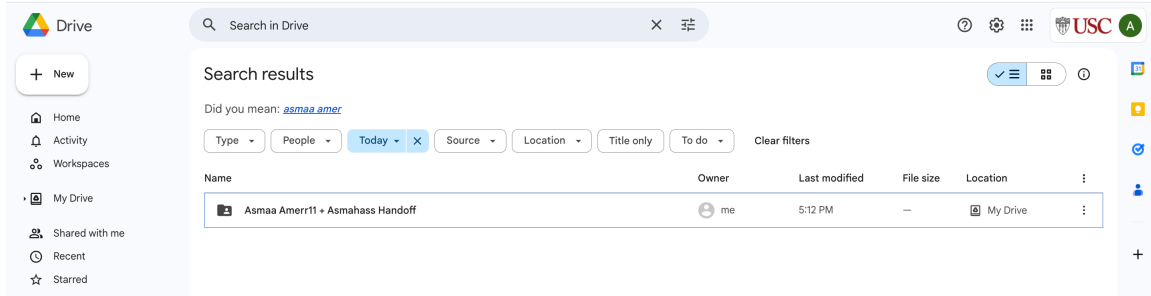
1. The script checks the event details and creates a Google Drive folder named:
Participant A + Participant B Handoff
2. The folder is shared with Participant A.

Step 4.2: When Participant B submits the form:

1. The script is triggered automatically via onFormSubmit.
2. It retrieves the form responses, formats them into a PDF file, and saves the file to the created folder.
3. Both Participant A and B receive email notifications with details about the saved file.

Step 5: Testing the Workflow

1. Create a test event in Google Calendar with at least two participants.
2. Check Google Drive to see if the correct handoff folder is created.



Apps Script Streamlining Operation

Deploy

Showing 27 executions over last 7 days | Show in real time:

Executions

+ Add a filter

Deployment	Function	Type	Start Time	Duration	Status
Head	checkForNewEvents	Time-Driven	Feb 7, 2025, 5:15:16 PM	1.024 s	Completed
Head	checkForNewEvents	Time-Driven	Feb 7, 2025, 5:14:16 PM	1.787 s	Completed
Head	checkForNewEvents	Time-Driven	Feb 7, 2025, 5:13:15 PM	1.572 s	Completed
Head	checkForNewEvents	Time-Driven	Feb 7, 2025, 5:12:18 PM	1.207 s	Completed
Head	checkForNewEvents	Time-Driven	Feb 7, 2025, 5:11:16 PM	3.985 s	Completed

Cloud logs

Time	Log Level	Message
Feb 7, 2025, 5:11:17 PM	Info	Checking for new events...
Feb 7, 2025, 5:11:18 PM	Info	Found 1 new event(s).
Feb 7, 2025, 5:11:20 PM	Info	Folder "Asmaa Amerr11 + Asmahass Handoff" created and shared with Asmaa Amerr11
Feb 7, 2025, 5:11:20 PM	Info	Next sync token saved.
Feb 7, 2025, 5:11:20 PM	Info	New events processed, and folders created.

REFRESH

3. Check access to Google Frome and Submit the form using the public link.

Apps Script Streamlining Operation

Deploy

Files

Code.gs

```

159 // Retrieve the last saved sync token
160 getNextSyncToken() {
161   const token = this.props.getProperty(this.key.nextSyncToken);
162   if (!token) Logger.log("No sync token found.");
163   return token;
164 }
165
166 // // Manually Create a trigger to monitor form submissions
167 function createFormTrigger() {
168   const formId = "14r1sUPTDv6xamXvU1upm2jsSA2Ik12iwlYmYRhjPbI"; // Replace with your form ID
169   const form = FormApp.openById(formId);
170
171   // Create a trigger to run on form submission
172   ScriptApp.newTrigger("handleFormSubmission")
173     .forForm(form)
174     .onFormSubmit()
175     .create();
176   Logger.log("Form submission trigger successfully created.");
177 }
178
179 // Manually triggered functions for initial setup and testing
180 function initialSync() {
181
182

```

Run

testFormAccess

Execution log

Execution log

5:20:03 PM	Notice	Execution started
5:20:04 PM	Info	Successfully accessed form: Handoff Forme
5:20:04 PM	Notice	Execution completed

docs.google.com/forms/d/e/1FAIpQLSdsz07iviOZvV5qx_xtDiA3vS6DMWwfM-N68YVvWIBAhEyFYQ/viewform



Handoff Forme

Lorem ipsum dolor sit amet, consectetur adipiscing elit.

asmahass@usc.edu [Switch account](#)

Not shared [Draft saved](#)

* Indicates required question

Name *

AsmaUSC

Email *

asmahass@usc.edu

docs.google.com/forms/u/0/d/e/1FAIpQLSdsz07iviOZvV5qx_xtDiA3vS6DMWwfM-N68YVvWIBAhEyFYQ/formResponse



Handoff Forme

Your response has been recorded.

This form was created inside of University of Southern California.
Does this form look suspicious? [Report](#)

Google Forms

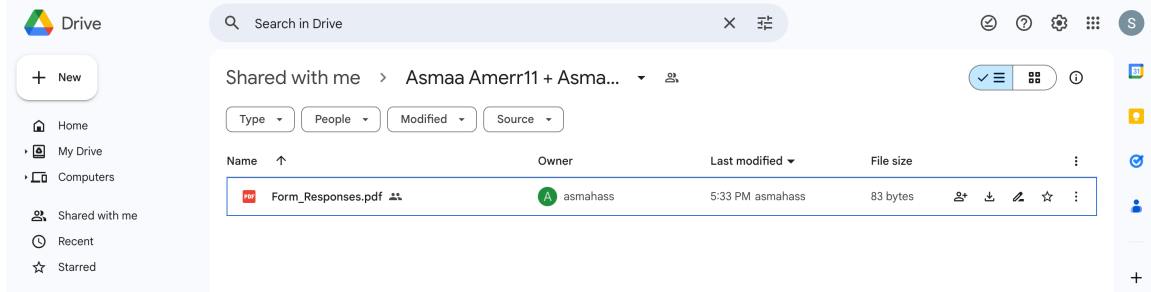
4. Confirm that:

- The PDF file with form responses is saved in the correct folder.

Executions Showing 44 executions over last 7 days | Show in real time: ☒

[+ Add a filter](#)

Deployment	Function	Type	Start Time	Duration	Status
Head	checkForNewEvents	Time-Driven	Feb 7, 2025, 5:28:17 PM	1.344 s	Completed
Head	checkForNewEvents	Time-Driven	Feb 7, 2025, 5:27:15 PM	1.155 s	Completed
Head	handleFormSubmission	Trigger	Feb 7, 2025, 5:27:01 PM	3.192 s	Completed



- Both participants receive email notifications.



Step 6: Troubleshooting

1. If the folder isn't created: Check the execution logs for errors in `initialSync` or `checkForNewEvents`.
2. If the form submission trigger fails: Ensure the form ID is correct and the form is shared with the Apps Script account.